

# VICTOR IBHAFIDON

## Machine learning Engineer

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### PROFESSIONAL SUMMARY

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My approach to software engineering is grounded in the belief that good software is built through thoughtful design, sound engineering, and continuous iteration. I've worked on AI/ML solutions across software engineering, data, security, and cloud, gaining experience through building, iterating, and learning from challenges along the way, while balancing technical requirements with broader business needs. My philosophy is simple: I don't just want something that works in a demo. I want to understand what can go wrong, prove it works, and build it properly. I own infrastructure as well as application code, using Terraform-managed AWS environments, Docker and Kubernetes delivered through CI/CD.

### TECHNICAL SKILLS

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**AI & Machine Learning:** RAG, PyTorch, Scikit-learn, LLMs, Generative AI, AI Agents, Multi-Agent Systems, LangGraph, MCP, JSON-RPC, Tool Calling, GraphRAG, Prompt Engineering, AI Evaluation, Transformers, Computer Vision, NLP

**Cloud & DevOps:** Docker, Kubernetes, AWS (EC2, S3, Lambda, IAM, EKS), Microsoft Azure, Azure OpenAI, GitHub Actions, CI/CD, Linux, Nginx, OpenTelemetry, Prometheus, Grafana

**Programming:** Python, TypeScript, JavaScript, SQL, Bash

**Security & AI Governance:** Responsible AI, AI Security, AI Governance, Authentication, Authorization, RBAC, API Security, OWASP Top 10, Prompt Injection Mitigation, AI Safety, GDPR

**MLOps & Model Operations:** MLflow, Model Training, Model Evaluation, Model Monitoring, Drift Detection, Experiment Tracking, Model Deployment

**Backend & Data:** FastAPI, Django, REST APIs, WebSockets, Microservices, PostgreSQL, MySQL, MongoDB, Redis, Neo4j, pgvector, Vector Search, Data Pipelines, ETL

### PROFESSIONAL EXPERIENCE

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#### Software Engineer (AI/ML), Bastion Shield Technologies

London, United Kingdom | April 2026 – Present

- Built Bulwark, a multi-tenant AI agent governance platform that gives organisations control and visibility over AI agents. Designed and implemented agent identity and lifecycle management, policy enforcement, human oversight workflows, and detection of unregistered agents, orphaned ownership, and stale credentials.
- Developed behavioural anomaly detection combining statistical methods with usage forecasting to identify unusual agent activity. The system achieved a 0.981 ROC-AUC against an Isolation Forest baseline and was integrated into the TypeScript runtime for real-time scoring.
- Engineered the backend in Python, TypeScript, and PostgreSQL, implementing RBAC, server-side tenant isolation, secure authentication, API-key management, and an SDK for external teams to register agents and stream policy decisions. Designed tamper-evident audit logging with offline verification, strengthened data privacy controls, and remediated four production vulnerabilities, including a session-token forgery path.
- Owned the cloud infrastructure using Terraform-managed AWS environments across development, staging, and production. Containerised the platform with Docker and delivered it through GitHub Actions CI/CD, while working on distributed system patterns including queues, event streaming, caching, and horizontal scaling to improve performance and latency.

#### Full Stack Software Engineer, Basketball Nxtion

London, United Kingdom | February 2026 – April 2026

- Developed the MVP of Pxyground, a mobile application connecting basketball players, creators, and sports communities, contributing across product design, application architecture, backend development, APIs, authentication, and user workflows.
- Designed the PostgreSQL data architecture and analytics pipelines for the MVP, structuring user, activity, engagement, and platform data to support data ingestion, transformation, analysis, and business intelligence. Built data workflows enabling insights into platform performance, user behaviour, engagement, and product adoption.
- Designed and integrated an AI architecture connecting LLM/ML services with PostgreSQL data, backend APIs, and application business logic. Built workflows for personalised recommendations, intelligent user matching, and content discovery, using user and platform data to generate relevant results, improve engagement, and automate discovery workflows.
- Deployed and maintained the MVP infrastructure on AWS, implementing Docker, CI/CD, testing, rollbacks, monitoring, and third-party integrations while collaborating with frontend engineers and stakeholders throughout the software development lifecycle from requirements and development through testing, deployment, and iteration.

#### Full Stack Python Engineer Intern, CyberAI Technologies

London, United Kingdom | September 2025 – November 2025

- Collaborated with senior AI engineers to develop Surfyn AI, a conversational AI customer support platform enabling businesses to deploy knowledge-based AI assistants grounded in websites and documents to automate customer interactions and provide relevant, context-aware responses.
- Contributed to the development of RAG and conversational AI workflows using Python and Django, working on document ingestion, chunking, embeddings, vector-based retrieval, prompt construction, conversation context, REST APIs, and LLM integrations while learning how knowledge-based AI systems are designed and evaluated.
- Developed backend functionality and AI workflow improvements, working on API development, debugging, testing, performance optimisation, knowledge retrieval, and integration between AI services and the Django application.
- Worked with the AI Security Engineering team on AI evaluation, testing, safety, and deployment workflows, helping investigate retrieval quality, response relevance, hallucination risks, prompt injection, sensitive data handling, logging, and application reliability across production AI systems.

## SELECTED PROJECTS

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**Portivo Estate Suite (AI Powered PropTech Operating System):** Approached Portivo from a consulting and product-engineering perspective, working with estate managers to understand how AI could be integrated into existing property operations to reduce administrative workload and improve lettings, sales, maintenance, compliance and reporting. Worked directly with users through discovery, workflow design, testing and implementation, translating feedback into product requirements, documenting APIs and operational processes, validating end-to-end workflows and supporting customer configuration and adoption. Designed the system architecture and took ownership of end-to-end development across product analysis, UX/UI, frontend, backend, AI/ML, data engineering, cloud, security, QA, AI governance and integrations. Built the multi-tenant application with React, TypeScript, Next.js and Tailwind, developing dashboards, property management, CRM, reporting and administrative workflows, backed by Python, FastAPI and SQLAlchemy for business logic and APIs. Designed the PostgreSQL data model with tenant-scoped queries and Alembic migrations against a live customer database, using Redis and Celery for caching, asynchronous processing and scheduled jobs. Built the AI architecture using LangGraph, Anthropic Models, RAG, OCR, chunking, embeddings and pgvector, with an orchestrator delegating typed tasks to specialist agents for document retrieval, lease abstraction, arrears detection, compliance tracking, maintenance triage and outbound sales prospecting. Implemented MCP servers and scoped credentials to give agents controlled access to CRM, email and document tools rather than creating bespoke integrations. Evaluated AI performance using labelled datasets for groundedness and extraction accuracy, with Langfuse for tracing, schema-validated outputs, confidence thresholds and audit trails for agent actions. Applied RBAC, tenant isolation, scoped permissions, security controls and AI governance to protect customer data and control agent behaviour, including prompt-injection and data-leakage risks. Containerised and deployed the platform to the cloud using Docker and GitHub Actions CI/CD, while designing API integrations, technical documentation and implementation workflows to support enterprise adoption. **Live in production at:** [portivoestatesuite.com](https://portivoestatesuite.com)

**Autonomous Energy Grid and Smart Battery Management:** A production-ready prototype of an AI-enabled platform that optimises energy distribution across a regional smart grid by integrating and managing thousands of Raspberry Pi-based smart batteries, so distributed storage can be coordinated centrally instead of managed device by device. Built as an end-to-end ML and MLOps system covering feature engineering, model training and evaluation, deployment and drift monitoring, using MLflow, Docker, Kubernetes and GitHub Actions with Prometheus, Grafana and OpenTelemetry observability. GitHub: [github.com/Web8080/Energy-Grid-and-Battery-Management-System](https://github.com/Web8080/Energy-Grid-and-Battery-Management-System)

## EDUCATION

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**MSc Data Science**, Nottingham Trent University, United Kingdom (2023 – 2024)

**BSc Computer Science**, Ambrose Alli University, Nigeria (2016 – 2021)